Energy-selective four-time correlations in dissipative Majorana systems

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We identify a four-time fermion correlation that selects energy relaxation within a controlled finite-time window in finite Majorana systems. For homogeneous interactions and uniform linear loss and gain, summing each of two short fermion pairs over all flavors selects the Hamiltonian at leading nontrivial order. After subtraction of the product of the within-pair correlations, equal pair durations ϵ produce an energy signal of order ϵ^2 and a complementary contribution of order ϵ^4 at fixed interpair delay. A norm bound controls this separation without resolving the complementary Liouvillian spectrum. Finite-system Sachdev–Ye–Kitaev calculations quantify the window and the accompanying decrease in signal strength. The complete finite-sample energy amplitude requires complex source and readout vertices, beyond the information fixed by the two-point fermion correlator. A mixed-interaction-order control identifies homogeneity as a condition for leading energy selection. Together, these results turn a known exact decay mode into a specified multitime correlation with controlled contamination and a source-dependent amplitude.

I. INTRODUCTION

Selecting a relaxation mode requires control of how it is excited and read out. The slowest Liouvillian mode can have little overlap with a chosen probe, while a faster mode can dominate the measured time window [1, 2]. An exact decay rate therefore becomes predictive for a concrete correlation only after its source, amplitude, and competing contributions have been specified.

Majorana systems with linear dissipation make this selection problem accessible at finite size. The dissipative Sachdev–Ye–Kitaev (SYK) model has been studied through its single-fermion Green's function and Liouvillian spectrum [3, 4]. For a Hamiltonian containing only q-body interactions, energy is an exact decay mode with rate $q\mu$ [5], illustrating the general relation between conserved quantities and simple dissipative decay [6]. Visibility and spectral ordering remain separate: the energy mode can be invisible to simple equal-time bilinears, and at $q = 4, N = 6$ the parity-dressed energy has rate 2μ, below the energy rate 4μ [5]. A probe of energy relaxation must therefore select its direction without assuming it is the slowest mode.

We construct such a probe from two short fermion pairs in a chronological four-time correlation. Complete flavor averaging makes the leading nontrivial part of each pair proportional to the Hamiltonian. The two pairs then act as an energy source and readout, with a finite-duration admixture whose relative size has a controlled bound. This gives an energy-selective window for an individual finite system, without a thermodynamic limit or disorder averaging. Computing the full source and readout amplitudes also identifies the information that a two-point correlator leaves undetermined.

The construction uses fermionic quantum regression to fix the signs between insertions [7, 8], within the Markovian regression framework [9] and its multitime extension [10]. Energy-vertex interpretations are established for nonalternating four-point correlations in closed, low-temperature SYK [11]. Here the finite-system dissipative setting permits exact energy projection together with finite-duration error control and a complete complex amplitude. The Supplemental Material gives the derivations [12].

II. A CORRELATION THAT SELECTS ENERGY RELAXATION

Consider even N Majoranas, $\{\psi_i, \psi_j\} = \delta_{ij}$, with a nonzero Hamiltonian

$$H = \sum_{i_1<\cdots<i_q} J_{i_1\cdots i_q}\psi_{i_1}\cdots\psi_{i_q}, \qquad L_i = \sqrt{\mu}\,\psi_i. \tag{1}$$

We take real couplings, $q \equiv 0 \pmod 4$, and $\mu > 0$. The Markovian evolution has Lindblad form [13, 14]. Each jump exchanges a fermion with the environment; equal rates leave the maximally mixed state $\rho_\infty = I/d$, $d = 2^{N/2}$, stationary. In terms of complex fermions, a pair of Majorana jumps describes balanced loss and gain. The analytic results allow arbitrary couplings in Eq. (1). The numerical examples use the Gaussian SYK ensemble [11, 15], with $\overline{J_I^2} = J^2/\binom{N-1}{q-1}$.

In the operator-size basis [16], uniform noise damps a Majorana string at a rate proportional to its length [17–19]. Since every term in H has the same length, the exact energy autocorrelation is

$$\langle H(g)H(0)\rangle_\infty = ve^{-q\mu g}, \qquad v = d^{-1}\operatorname{tr} H^2. \tag{2}$$

This established mode fixes the propagation between an energy source and readout. The four-time construction below supplies both from elementary fermion insertions.

Take four ordered times with consecutive separations $s, g, u > 0$, as in Fig. 1(a). We study

$$G_4(s,g,u) = \frac{1}{N^2}\sum_{i,j}\big\langle\psi_i(s+g+u)\psi_i(s+g) \times \psi_j(s)\psi_j(0)\big\rangle_\infty. \tag{3}$$

The sum includes $i = j$. Subtracting the product of the two within-pair correlations gives $G_{\text{pair}} = G_4 -$

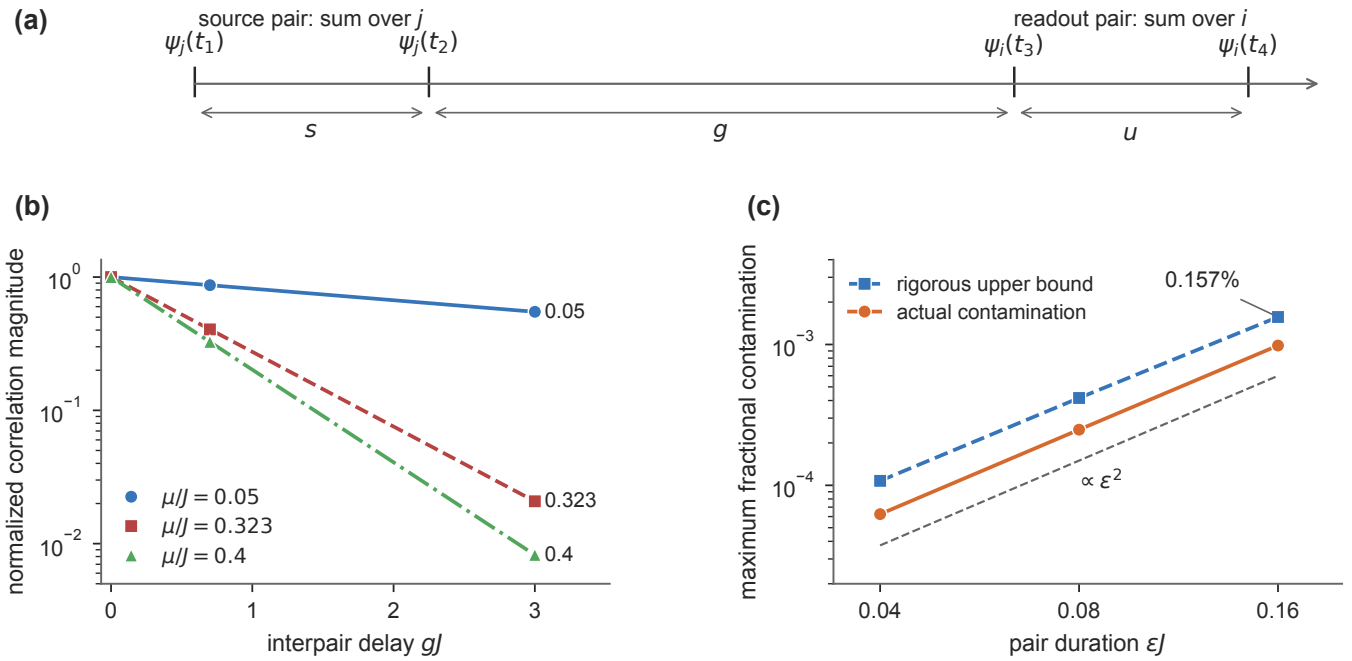


FIG. 1. Energy selection at finite pair duration. (a) Chronological insertions defining Eq. (3); the pair sums include every flavor combination. (b) Magnitude of $G_{\text{pair}}(\epsilon, g, \epsilon)$ normalized at $g = 0$, for one $N = 10, q = 4$ realization and $\epsilon J = 0.16$. Symbols are the three calculated delays; lines show $e^{-4\mu g}$ without a fitted rate. The middle noise value is $\mu/J = 0.3228079267$, rounded in the legend. (c) Largest relative contamination $|G_\perp|/|G_H|$ and largest rigorous bound among the 18 checked systems, at $gJ = 0.7$, where $G_H(s, g, u) = \mathcal{A}_H(s, u)e^{-q\mu g}$. The two maxima need not come from the same system. The gray dashed guide shows quadratic scaling, not a fitted exponent or a thermodynamic extrapolation.

$C_N(u)C_N(s)$, where $C_N(t) = N^{-1}\sum_i \langle \psi_i(t)\psi_i(0)\rangle_\infty$. This removes the stationary contribution between pairs; it is not the cumulant obtained by subtracting every Wick pairing. Fermionic regression is understood throughout. The observable is a multitime correlation, rather than the ordinary Hamiltonian linear susceptibility [20], which vanishes at ρ_∞.

Projecting the source and readout onto energy gives the exact decomposition

$$G_{\text{pair}}(s, g, u) = \mathcal{A}_H(s, u)e^{-q\mu g} + G_\perp(s, g, u). \quad (4)$$

The amplitude $\mathcal{A}_H$ is the product of the source and readout overlaps with energy. Pair dynamics can make it depend on μ, while all interpair-delay dependence of the energy contribution lies in the exponential in Eq. (4). The remainder $G_\perp$ contains the complementary directions, including any mode degenerate with the energy rate.

The full flavor sum converts a short pair into an energy vertex. Immediately after the first insertion, its short-time change contains the interactions involving that flavor. Summing over flavors counts every q-body term exactly q times, producing the total Hamiltonian with no component along another nonstationary direction. The second pair makes the same selection on readout. A single flavor probes a partial Hamiltonian; complete averaging in Eq. (3) is what makes the source and readout select total energy.

For equal short durations, $s = u = \epsilon$, the energy amplitude and the remainder obey

$$\mathcal{A}_H(\epsilon, \epsilon) = -\frac{q^2 v}{N^2}\epsilon^2 + O(\epsilon^3),$$
$$G_\perp(\epsilon, g, \epsilon) = O(\epsilon^4). \quad (5)$$

For fixed N and fixed g, this order separation gives quadratic suppression of relative contamination as the pairs are shortened. Energy requires one short-time interaction in each pair; a complementary contribution requires an additional order in both source and readout. The negative leading amplitude reflects the ordered correlation and is consistent with its being a product of vertices, not a probability or an absolute square.

III. FINITE DURATION, SIGNAL STRENGTH, AND THE OBSERVATION WINDOW

The energy-selection window can be bounded directly from the nonenergy components generated by the two pairs, without diagonalizing the complementary Liouvillian. The estimate applies the Hermitian-part contraction principle underlying spectral and transient-correlation bounds [21, 22]. Figure 1(b) illustrates the selected exponential in a finite SYK sample. The comparison tests the correlation within that window, inde-

pendently of which mode sets the full spectral gap. Supplemental Material, Sec. V gives the norm bound and its proof.

Finite-system calculations test both the size of the contamination and the bound controlling it. The set contains $q = 4$, $N = 6, 8, 10$, two realizations at each size, and three noise strengths $\mu/J = 0.05, 0.3228079267, 0.4$. At $gJ = 0.7$ and $\epsilon J = 0.16$, the relative-contamination bound is below 0.157% for all 18 systems, with a largest actual contamination of 0.0983%. Its decrease with pair duration in Fig. 1(c) supports the predicted order separation. These are maxima over the checked systems, not disorder confidence intervals. Direct comparisons of the four separate insertions with the source–readout construction, together with checks of the short-time coefficients, test the mechanism underlying the bound.

Shorter pairs improve selectivity while reducing the signal itself. Equation (5) shows that halving the pair duration reduces the leading energy signal by a factor of four, along with the quadratic improvement in relative contamination. For the chosen SYK normalization, $\overline{v} = NJ^2/(q2^q)$, so the leading disorder-averaged signal at fixed g is $-qJ^2\epsilon^2 e^{-q\mu g}/(N2^q)$. This is an averaged short-time coefficient; finite-duration self-averaging is not assumed. Ancilla protocols for controllable unitary dynamics reconstruct both real and imaginary multitime correlations [23]. A measurement advantage for the present dissipative probe would require a sampling analysis that includes this decreasing signal and the cost of reconstructing the ordered correlation.

The bound depends on the competition of decay rates: energy decays at $q\mu$, while a complementary even mode can decay more slowly. The relative-error bound can therefore grow with g, even as the absolute upper bound decreases. Shortening the pairs suppresses contamination at finite delay; it does not guarantee energy dominance at arbitrary late times. Extending the small-ϵ expansion uniformly in system size would require additional estimates.

IV. AMPLITUDE INFORMATION AND THE ORIGIN OF SELECTIVITY

The complete energy amplitude in Eq. (4) contains information beyond the decay rate and the two-point correlator. At finite pair duration, the source and readout have complex energy overlaps: their imaginary parts are fixed by the time derivative of the two-point fermion correlator, while their real parts generally survive in an individual sample. The leading real part is proportional to the third energy moment $d^{-1}\,\mathrm{tr}\,H^3$, which can be nonzero even for a pure quartic Hamiltonian. Recovering the full energy contribution therefore requires both parts of the vertices.

The finite-sample comparison in Fig. 2 quantifies this information gap for one $N = 8$ sample. At $\epsilon J = 1$, the complete energy amplitude is $-0.00838880 - 0.00051700i$,

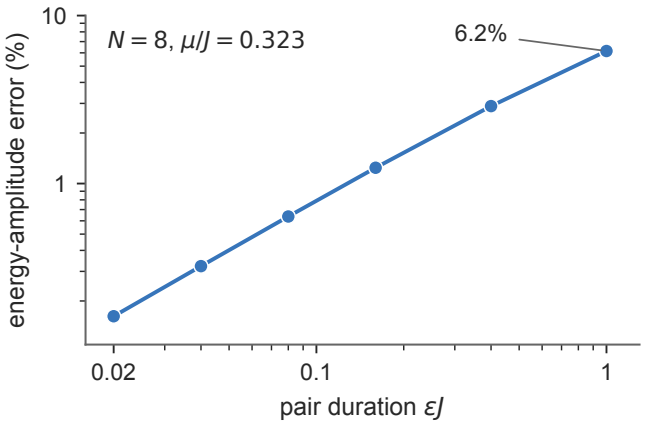


FIG. 2. Two-point information leaves part of the finite-sample energy amplitude undetermined. The plotted fractional discrepancy compares the complete $\mathcal{A}_H(\epsilon, \epsilon)$ with the result obtained by retaining only the imaginary energy vertices fixed by C_N. Data are for $N = 8, q = 4$, one realization, and $\mu/J = 0.3228079267$. Lines connect evaluated pair durations. This is an amplitude-reconstruction error within the energy contribution, distinct from the complementary-mode contamination in Fig. 1(c).

whereas retaining the part fixed by the two-point function gives -0.00839676, a complex-amplitude discrepancy of about 6.2%. This tests reconstruction of the isolated energy contribution, separately from contamination by other modes in the full four-point function. A disorder ensemble symmetric under $H \mapsto -H$ removes the mean real vertex but can retain its covariance in the mean energy amplitude. A product of averaged vertices therefore requires an additional self-averaging argument.

Mixing interaction orders tests the origin of selectivity directly. For $H = H_q + H_p$, the first short-pair contribution couples to $qH_q + pH_p$, so its unequal weights introduce a nonenergy component before interpair evolution begins. In a quartic $N = 8$ sample with an added term $0.25J\, i\psi_1\psi_2$, this component accounts for 2.39% of the squared norm of the initial size-weighted vertex. Total energy is then generally no longer a single-rate decay mode. The selection rule thus rests on homogeneous interaction order, uniform noise, and the full flavor sum; randomness and chaotic dynamics are not required.

V. DISCUSSION

The four-time construction makes energy relaxation visible by selecting its source and readout, independently of whether energy is the slowest mode. Shortening the pairs controls contamination and reduces the signal together, so the useful object is a finite observation window with a known amplitude. The complex vertices also separate knowledge of a mode and its rate from knowledge of its full contribution to a correlation: the latter retains source information absent from the two-point function.

This finite-system result complements the large-N collective description. General SYK four-point kernels and their sourced recursions are established [24, 25]. The Supplemental Material gives an explicit energy eigenfunction of the continuous singlet kernel and relates its shape to the imaginary energy vertex. This is an analytic consistency relation; obtaining the complete finite-duration residue requires the additional vertex information discussed above. Neither spectral completeness nor a thermodynamic gap transition follows from the relation, and loss of integrability in the relative-time representation does not remove the finite-system energy contribution.

The selected energy component, controlled finite-time remainder, and mixed-order breakdown together provide a benchmark for approximations to multitime open-system dynamics. They specify how an exact decay mode enters a concrete correlation, with its full source-dependent amplitude and the conditions for its selective observation.

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